

CLIENT API DOCUMENTATION

Google 360 Location API

Coordinate enrichment, Google Maps imagery and location metadata

Document item	Value
API version	1.1.0
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Audience	Client application, integration and support teams
Base URL	Deployment-specific; supplied by the hosting team

Purpose

This guide defines the client-facing HTTP contract for submitting one or more geographic coordinates and receiving Google Maps-derived imagery and location metadata. It covers request construction, response handling, field definitions and expected error behaviour.

Security boundary: Clients do not send or receive the server-side Google Maps credential. Network authentication, TLS and any gateway headers are deployment-specific and must be supplied by the hosting team.

Service overview

Google 360 is a synchronous HTTP API that accepts latitude and longitude values, enriches them through Google Maps services, and returns imagery plus address and place metadata. It does not provide a user interface and does not persist request or response data between calls.

Endpoint summary

Method	Path	Purpose	Success response
GET	/health	Service liveness check	JSON
GET	/api/get_data	Process one coordinate	ZIP or JSON
POST	/api/get_data	Process one coordinate or an uploaded file	ZIP or JSON

Base URL and access

All paths in this document are relative to the deployment base URL:

```
https://<client-approved-host>/<path>
```

- Use HTTPS in production.
- Apply gateway authentication headers exactly as provided by the hosting team.
- Do not include a Google Maps API key in client requests.

Quick-start examples

Single coordinate, ZIP response:

```
curl -OJ "https://<host>/api/get_data?lat=19.0760&lng=72.8777"
```

Bulk CSV upload, ZIP response:

```
curl -OJ -F "file=@coords.csv" "https://<host>/api/get_data"
```

Single coordinate, JSON response:

```
curl "https://<host>/api/get_data?lat=19.0760&lng=72.8777&format=json"
```

Health endpoint

GET /health

Returns a liveness response when the application process is available.

Item	Value
Method	GET
Path	/health
Parameters	None
Success status	200 OK
Content-Type	application/json

```
{"status": "ok"}
```

Scope: This is a liveness check only. It does not verify Google API availability, quotas or credentials.

Coordinate enrichment endpoints

GET /api/get_data

Processes one coordinate supplied through query parameters.

Parameter	Location	Type	Required	Validation / meaning
lat	Query	number	Yes	Latitude from -90 through 90
lng	Query	number	Yes	Longitude from -180 through 180
format	Query	string	No	zip (default) or json

```
GET /api/get_data?lat=19.0760&lng=72.8777&format=zip
```

POST /api/get_data

Processes either one coordinate or one uploaded bulk file. Supply one input mode per request.

Input mode	How to supply it	Content type	Typical use
Single coordinate	lat and lng query parameters	No request body required	Real-time lookup

Input mode	How to supply it	Content type	Typical use
Bulk file	Multipart field named file	multipart/form-data	Multiple locations

POST, single coordinate:

```
curl -OJ -X POST "https://<host>/api/get_data?lat=19.0760&lng=72.8777"
```

POST, bulk file with JSON response:

```
curl -F "file=@coords.csv" "https://<host>/api/get_data?format=json"
```

Bulk-file contract

Requirement	Accepted value	Notes
File type	CSV or XLSX	CSV is recommended. Legacy XLS availability depends on the deployed parser.
Header row	Required	The first row must contain column names.
Latitude column	lat or latitude	Case-insensitive; surrounding spaces are ignored.
Longitude column	lng, long or longitude	Case-insensitive; surrounding spaces are ignored.
Additional columns	Allowed	Additional input columns are ignored by the API.
Maximum rows	500 by default	The deployment owner can configure a different limit.
Excel worksheet	First worksheet	Only the first worksheet is read.

Example CSV:

```
lat,lng
19.076090,72.877426
28.613939,77.209023
12.971599,77.594566
```

Validation: Blank, non-numeric and out-of-range file rows are reported individually and are not sent to Google. The request can still return 200 when other rows succeed.

Response formats

Both enrichment endpoints accept format=zip or format=json. When format is omitted, ZIP is returned.

Format	Content-Type	Recommended use	Important behaviour
zip	application/zip	Downloads and bulk integrations	Most compact; includes summary.csv and summary.json

Format	Content-Type	Recommended use	Important behaviour
json	application/json	Direct system-to-system filing	Files are Base64 encoded and therefore larger

ZIP response

The response filename follows this pattern:

```
{ZIP_NAME_PREFIX}_{UTC timestamp}_{8-character identifier}.zip
```

A successful two-row archive can contain:

```
1/map.png
1/street.jpg
1/360.html
2/map.png
2/360.html
summary.csv
summary.json
```

- Location folders are numbered from 1 in submitted row order.
- street.jpg is omitted when no Street View imagery is available.
- Optional fixed-bearing images use names such as street_000.jpg and street_090.jpg.
- 360.html is an interactive web page, not an image; it requires browser and network access.

JSON response

The JSON representation includes the same result metadata plus a files object. Each file value is a Base64 string without a data-URL prefix.

```
{
  "count": 1,
  "generated_utc": "2026-08-27T10:30:00+00:00",
  "locations": [{
    "folder": "1",
    "lat": 19.076,
    "lng": 72.8777,
    "status": "ok",
    "note": "",
    "address": "Example address",
    "static_map": "1/map.png",
    "street_img": "1/street.jpg",
    "street360": "1/360.html",
    "files": {
      "map.png": "<base64>",
      "street.jpg": "<base64>",
      "360.html": "<base64>"
    }
  }]
}
```

Bulk recommendation: Prefer ZIP for large batches. Base64 increases file payload size by approximately one third before JSON overhead.

Per-location result fields

The following fields appear in summary.csv, summary.json and JSON locations entries.

Field	Type	Description
folder	string	One-based location number and ZIP folder name.
lat	number/string	Validated latitude. Invalid rows may contain an empty value.
lng	number/string	Validated longitude. Invalid rows may contain an empty value.
status	string	ok, partial or error.
note	string	Semicolon-separated explanation of missing data or processing failures.
address	string	Formatted address from Google; blank when unavailable.
place_name	string	Google place name when available.
business_status	string	Google business state, such as OPERATIONAL or a closed state.
types	string	Pipe-separated Google place types.
place_id	string	Google place identifier returned by reverse geocoding.
street_view_status	string	Street View metadata result, commonly OK or ZERO_RESULTS.
static_map	string	Path to map.png in the ZIP; blank if generation failed.
street_img	string	Path to street.jpg; blank if unavailable or failed.
street_extra	string	Pipe-separated paths to optional fixed-bearing Street View images.
street360	string	Path to the interactive 360.html file.
files	object	JSON mode only: filename-to-Base64 mapping for generated files.

Result status meanings

Status	Meaning	Client action
ok	All expected processing steps completed.	Store or present the returned assets.
partial	At least one asset or lookup failed, but one or more files were produced.	Use available files and inspect note before completing the workflow.

Status	Meaning	Client action
error	The row produced no usable files, commonly because the coordinate was invalid.	Do not treat the location as successfully verified; inspect note and correct or retry.

Important: HTTP 200 does not mean every submitted row succeeded. Always inspect status and note for each location.

HTTP errors and recovery

HTTP status	Condition	Expected client handling
200	Request completed; rows may still be partial or error	Parse every result row and inspect status and note.
400	Missing input, unsupported file type, unreadable file, missing columns, empty data or row limit exceeded	Correct the request. Do not retry unchanged.
401 / 403	Possible gateway authentication or authorization rejection	Verify deployment credentials and access policy.
413	Possible gateway upload-size rejection	Reduce or split the upload.
422	Query parameter failed type, range or format validation	Correct lat, lng or format.
500	Unexpected application failure	Record the response and contact support; retry only under the agreed policy.
502 / 503 / 504	Possible proxy, service or timeout failure	Retry with backoff if permitted; avoid rapid repeated requests.

400 response example

```
{"detail": "Provide either an uploaded file or both 'lat' and 'lng' query parameters."}
```

422 response shape

```
{
  "detail": [{
    "type": "less_than_equal",
    "loc": ["query", "lat"],
    "msg": "Input should be less than or equal to 90",
    "input": "999"
  }]
}
```

Client integration guidance

Recommended processing sequence

- Obtain the deployment base URL and gateway authentication requirements.
- Submit one coordinate or one file per request.
- Confirm the HTTP response status and Content-Type.
- For ZIP, extract summary.json or summary.csv before processing asset folders.
- For JSON, iterate locations and Base64-decode each files entry using its filename.
- Inspect every location status and note before recording the integration outcome.
- Persist the original row-to-folder mapping for audit and reconciliation.

Python ZIP example

```
import io
import json
import zipfile
import requests

base_url = "https://<host>"
with open("coords.csv", "rb") as source:
    response = requests.post(
        f"{base_url}/api/get_data",
        files={"file": ("coords.csv", source, "text/csv")},
        params={"format": "zip"},
        timeout=180,
    )
    response.raise_for_status()
with zipfile.ZipFile(io.BytesIO(response.content)) as archive:
    results = json.loads(archive.read("summary.json"))
    for location in results:
        print(location["folder"], location["status"], location["note"])
```

Operational considerations

- Requests are synchronous. Use a client timeout that accommodates the agreed bulk size and gateway timeout.
- Split files that exceed the configured row limit; preserve an external batch identifier for reconciliation.
- Do not assume street.jpg exists. Street View coverage varies by location.
- Treat Google-supplied business metadata as supporting information rather than definitive evidence of current business operation.
- Do not parse visual files from their binary content when the response summary already supplies canonical paths.

- Avoid logging full Base64 payloads, complete addresses or coordinate-bearing URLs unless approved by the client data policy.

Interactive API definition

When enabled by the hosting team, the deployment exposes:

Path	Purpose
/docs	Swagger UI for interactive endpoint inspection and testing
/openapi.json	Machine-readable OpenAPI schema

Environment control: Access to interactive documentation may be restricted or disabled by the production gateway.

Integration readiness checklist

- Base URL received and reachable over HTTPS.
- Gateway authentication mechanism confirmed.
- Single-coordinate ZIP and JSON calls tested.
- Representative CSV or XLSX file accepted.
- ZIP summary parsing implemented before asset handling.
- Partial and error rows handled explicitly.
- Missing Street View image handled without failing the record.
- Client timeout, retry and maximum batch size agreed.
- Coordinates, addresses and images handled under the approved data-retention policy.

Contract authority: This document describes application version 1.1.0. Deployment-specific network, authentication, upload-size and timeout controls take precedence where they are stricter.